Noise-Induced Synchrony and the Effects on Spike Pattern Length in a Spiking Neural Network

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Abstract

The underlying coding mechanims of information processing in the brain are of high relevance in biological and technical research. Two main electrical coding schemes have been identified in neurons, rate code and temporal code. Rate code has received most attention in previous research because of strong and easily identifiable event-correlation results. In recent years, temporal code has regained interest, specifically on the topic of synchrony. Advantages provided by synchrony code include increased temporal precision and shorter timescales, signal processing of distributed information, and enhanced signal summation. The impact of synchrony code can be demonstrated by analyzing spike pattern lengths in a laterally connected spiking neural network. Increased synchrony is expected to shorten pattern length due to parallelization of spiking processes. In this experiment, the effects of noise-induced synchrony on spike pattern length in a spiking neural network were evaluated. The model consisted of a single recurrent layer representing a small laterally connected neuron cluster. Synchrony was found to decrease spike pattern length and thus promote efficient information processing for distributed signals.

1 Introduction

With the ambition to better understand how a complex system like the human brain coordinates around 86 billion processing units [11] to perform an estimate of 6000 thoughts per day [32], neural coding mechanisms are studied extensively. Previous work has revealed detailed insights on the physical structure of the brain as well as lower order mechanisms, but understanding how neural mechanisms lead to the thoughts we experience remains a challenge [13]. Different types of neural coding have been detected and researched, with the focus on rate code, because of its consistency in results and easy detection [13, 23, 36]. Rate-based code conveys information by relating the average pre-synaptic firing frequency to post-synaptic output, where higher pre-synaptic spike counts correlate with stronger or more likely post-synaptic signals. Research on other forms of neural coding is still sparse, because rate code has proven to be reliable, comprehensible, reproducible and robust in its results [23]. Temporal code has received less attention in previous research due to difficulties in simultaneously recording the activity of multiple neurons, and major doubts about its relevance. Specifically synchrony code, a sub-type of temporal code, has received strong criticism. Neural synchrony can be described as the firing of different neurons at the same time or with a consistent offset. Recordings of neural synchrony in experimental research are abundant, however the phenomenon has commonly been reduced to a mere by-product, disregarding its potential as a coding mechanism [14, 29, 36]. Despite the challenges, synchrony code has recently regained interest.

1.1 Synchrony Code

This type of temporal code works through a mechanism called *coincidence detection*, which attributes value to correlations between spike timings of different neurons, with promising theoretical advantages [13].

In comparison to rate code, synchrony code cannot be evaluated on the basis of singular neurons, but relies on multicell analysis, i.e. the recording and evaluation of different spiking neurons simultaneuously [29]. This creates the opportunity to relate spatially distributed signals to coherent patterns of functional importance. It may serve to solve the binding problem that rate code faces, due to it being limited to local operations [13, 29]. Spatial distribution of related signals also drastically increases the overall impact of information exchange on the whole neuronal system: With rate code operating on a singular cell level, one information unit is unable to influence the whole system when its information changes, due to longer pathways between singular neurons. In contrast, if information is encoded in multiple, spatially distributed cells, overall impact is increased through shorter pathways and combined activations. This enables encoding and decoding at a higher grain of neuronal representation, but also allows for more efficient code forwarding from spatially distributed, parallel processing [13, 36].

Not only does synchrony code lead to increased efficiency by greater spatial coverage, it can also operate on very fast timescales. While rate code relies on temporal summation, which works on timescales with the same order of magnitude as the average interspike interval (ISI) [13, 15], coincidence detection can process information on a few millisecond timescale, which is often only a fraction of the average ISI [13]. This leads to a substantial increase in processing speed. A further increase in efficiency through coincidence detection emerges from the nature of synchronized spikes, which have the potential to increase saliency through spatial summation and backpropagating dendrites [29]: For the same amount of absolute spikes, a stronger and more immediate response can be evoked from synchronous spatial summation compared to temporal summation. When temporal summation is not , post-synaptic spikes are delayed in time due to the averaging process. Furthermore, in the case of low spike frequency, the voltage may remain below the threshold entirely [15, 29]. In summary, synchrony code allows for increased efficiency at the temporal and spatial level, which further allows for important metabolic advantages [36].

With the many benefits of synchrony coding, it would be highly unlikely that the nervous system and the brain do not harness it at all. Nonetheless, temporal code does not come without downsides, which is why most of recent research suggests a combination of rate code and synchrony code, as well as other potential coding schemes to be the most plausible and efficient [13, 29, 36].

1.2 Neural Noise

Neural noise is a result of the neuronal network complexity on a molecular level. Ion channel activity, protein production and decomposition, as well as synaptic input artifacts are non-deterministic processes that cause irregular membrane voltage fluctuations [24]. In contrast to prior beliefs, recent studies have shown that neural noise is not merely a by-product that the system is built to work around, but that it may be of functional biological relevance [7, 17, 24]. Furthermore, neural noise was found to support or cause the emergence of synchrony, due to a phenomenon called stochastic resonance [17, 24, 26]. Stochastic resonance can occur if neurons receive an oscillatory drive as well as a non-zero amount of noise, which induces synchrony by shifting the activity phases of different involved neurons. It was also found that in non-linear systems, similar behavior can be induced even without a continuous external oscillatory drive. In this case, increased noise amplitude decreases the variation coefficient, leading to an increase in spike correlations. This phenomenon is referred to as *coherence resonance* [7, 17, 24].

1.3 Synchrony in Dynamical Systems Theory

Another perspective in favor of investigating noise effects on synchrony is research on modeling the brain as a complex dynamical system. Eggermont [6] suggests that the brain possesses key attributes of complex dynamical systems, specifically immense complexity and sensitivity to small changes. Dynamical systems theory offers a potential solution to the question of how small changes on the molecular level could lead to big overall changes manifesting in cognitive states. If we assume the brain to be a complex dynamical system, single-cell activity becomes much less relevant than multi-cell coding, because the functioning of a complex dynamical system relies heavily on its ability to perform global transitions. When modeling the brain as a dynamical system, such transitions would be carried out by neuronal activity, with synchrony as a promising candidate to initiate those [6]. Tsuda [33] argues that noise is a relevant component in modeling the brain as a complex dynamical system. He distinguishes between dendritic and synaptic noise, the former defined as spontaneous emissions which may influence subthreshold behavior, but too weak to activate a postsynaptic spike, whereas the latter is described as stimulus-induced noise, potentially resulting in post-synaptic spikes. Tsuda [33] also mentions the theory of noise to induce order despite its chaotic behavior, which drives the hypothesis in this paper.

1.4 Hypothesis

This paper aims to provide further insights into the effect of synchrony on temporal efficiency and its relevance as a coding mechanism. In addition, the experiment is supposed to demonstrate that synchrony does not necessarily rely on previous synchronous input, but can emerge from an optimal non-zero amount of noise, which is a biologically plausible initial state in neurons [2, 7, 34].

The proposed hypothesis is that (1) noise alone can induce synchrony in a laterally connected spiking neural network and (2) this synchrony is able to shorten spike pattern length. The proposed underlying mechanism is parallelization of processes: Assuming that a pattern running though the network structure induced by only one initial spike will perform one step after another until the pattern is finished, whereas the same pattern induced simultaneously at different points will be carried out in parallel, thus runtime decreases. Consequently the overall pattern length is shortened, leading to increased temporal efficiency.

2 Related Work

Theoretical arguments for the interest in synchrony were laid out in the introduction, however, experimental evidence on biological plausibility is crucial to justify further scientific investigation. The main criticism research on synchrony code receives is the possibility of it being an epiphenomenon without any efficacy [29]. To counter this criticism, new research has to prove that synchrony has functional importance by providing evidence of synchrony correlating with behavior independent of modulations in firing rate. Different in vivo experiments have shown the functional role of synchrony in attention and arousal states, cognitive tasks, for perception, expectancy and sensory mechanisms [29].

Synchrony has shown to be of functional significance across different sensory domains. Olfactory pathways in locusts were found to harness synchrony as a coding scheme for odor distinction. When synchrony was disrupted, odor selectivity in some neurons decreased and hence their discrimination ability was impaired [16]. A similar experiment with additional behavioral testing in bees confirmed the hypothesis that oscillatory synchronization in neurons at an early processing stage of the olfactory pathway codes for odor specificity. When synchronization of those neurons was inhibited, neurons that were activated later in the chain did not receive the input they needed for discrimination. Most importantly, this result was observed without significant changes in the firing rate of the desynchronized neurons, concluding functional importance of synchrony [30].

Evidence for the role of synchrony in perception comes from a study of cats with strabismic amblyopia. This is a condition where in order to avoid double vision, one eye becomes the dominant source for visual information, and the other eye develops perceptual deficits due to reduced usage. Synchrony patterns of neuron pairs each corresponding to one of the eyes were compared. It was found that neuron pairs connected to the normal eye showed significantly higher synchronization than pairs of the non-dominant eye. The observed effect was even stronger when cats were shown stimuli of different gratings, namely a coarse grating that could be identified reliably with both eyes, and a fine grating that was perceptible only with the normally functioning eye. For the normal eye, with finer gratings the response amplitude of neurons decreased, while synchronization remained stable. For the amblyopic eye, response amplitude as well as synchrony decreased. These results underline predictions that conditions where the amblyopic eye is at an even stronger disadvantage lead to even weaker synchronization. It has to be emphasized that the change in synchrony was not accompanied by firing rate modulations, which suggests that decreased synchronization is functionally correlated with impaired perception [28].

A similar study investigated cats with interocular rivalry, a condition where one eye acts dominant over the other at every point in time, with dominance switching between eyes, but one eye usually favored. The cats were shown either congruent stimuli, where both eyes were presented with the same image, or incongruent stimuli, where each eye received a different image. For the incongruent condition, synchrony in neuron pairs corresponding to the dominant eye increased, while synchrony in neurons of the non-dominant eye decreased. In addition to synchrony, firing rates of those same neurons were analyzed and displayed no significant changes when switching from the congruent to the incongruent stimulus. These findings of significant reduction in synchrony for the perceptually impaired eye, independently of firing rate, support the theory that synchrony code is used for perceptual processing in the brain. Results may imply that synchrony plays a role in attentional mechanisms [8].

Further evidence for the importance of synchrony in cognitive processes can be derived when looking at the brain structes where synchrony is found. Higher order cortical areas show increased synchrony compared to lower-order cortical areas [29, 36]. This synchrony gradient correlates with the degree of lateral interconnectivity, where the more strongly interconnected higher order cortical areas appear to utilize synchrony code more often than lower-order cortical areas, which are less densely laterally connected and presumed to mostly harness rate code. This makes sense considering the nature of synchrony code, which is reliant on strong internal communication for coordination of spike times, whereas rate code is more effective in detecting feedforward drive, as it cannot interpret distributed information [36]. These findings are in line with the observations that synchrony connects behaviorally to attention and perception, with final processing stages taking place in higher order cortical areas [29].

The role of synchrony was also investigated in the somatosensory and motor cortex. Synchronization was found to be greater in exploratory and self-organized movement tasks, rather than constrained and directed tasks [19]. A specific correlation to movement execution was not found so far; synchrony was associated with the movement preparation period or the anticipation of movement, but was found to dissolve right before the onset of movement [5, 19, 27]. Importantly, emergence of synchrony in the movement anticipation period was independent of firing rate changes [27]. These findings suggest that synchrony is not necessarily used to code for actual movement execution, but rather supports hypotheses of synchrony as a code for cognition, specifically perception and attention processes, such as binding of perceptual features or the mental separation of foreground and background [5, 19, 27]. De Oliveira et al. derive a potential feature binding mechanism from their results showing that most synchronized neuron pairs in their experiment had the same stimulus preference [5]. Murthy and Fetz proposed that an attentional mechanism could be carried out through synchrony code by simultaneously pushing assemblies of neurons that are attending to the same stimulus closer to firing threshold [19].

While in vivo experiments are necessary to prove biological plausibility, studying the theoretical potentials of synchrony in computational models is another crucial field of research. The field has revealed the potential of synchrony to provide a mechanism that can store familiarity and memory information, where the structure of connectivity in a neuronal network can represent known patterns [14, 36]. Memory by synchrony operates through Hebbian plasticity, a long-term potentiation mechanism where neurons that fire together repeatedly strengthen their connections [34, 36]. Experiments like these not only enable future research to form hypotheses of potential biological mechanisms to then be tested in vivo, they also support technological progress and thus stay highly relevant.

In summary, in vivo experiments on the one hand have gathered evidence for the functional role of synchrony in some areas of sensory processing, as well as for cognitive functions. More specifically, the type of cognitive fucntions that harness synchrony inlcude perception, attention and grouping of stimuli. On the other hand, experiments that test synchrony in computational models support this process and provide valuable insights about this coding scheme from a different angle.

3 The Model

3.1 Layer Setup and Recurrency

To examine how sub-threshold noise induces and affects synchrony, a spiking neural network (SNN) model was set up with a single recurrent layer. Recurrent layers as well as spiking neurons are not the standard procedure in current Artificial Neural Network (ANN) research, which SNNs are a sub-type of [3, 18]. Many ANN architectures rely on non-recurrent feedforward layers and the majority of artificial neurons utilize continuous activation functions at their synapses [34]. Recurrent ANN layers are commonly used for sequential learning that integrate past input into the next step, however randomly interconnected singular layers as used in this model are uncommon. The recurrent layer in this model mimics a very simple patch of lateral connections with a certain connectivity probability, which is a structure shown to facilitate synchrony. The mechanism enabling synchrony in such networks is a recurring communication among neurons, leading to a self-organized adaptation to each other's patterns through cross-activation [22, 36]. The choice of one singular layer over multiple connected recurrent layers, as would be more biologically plausible, was made to keep results and evaluation simplistic and to reduce computational load to a minimum.

3.2 The Artificial Neuron

Neurons with continuous activation functions have mathematical advantages: due to their comparably simple derivatives, backpropagation using gradient descent, which is the current dominating learning algorithm for ANNs, can easily be implemented [4, 34]. Spiking neural networks use a different type of integration at the synapse, where inputs sum up as binary input weights and create binary outputs, instead of continuous input weights integrated under a continuous function. Those binary outputs of spiking artificial neurons relfect more closely, but not completely, how biological neurons integrate. Very simply put, they communicate with the generation of a spike or no spike resulting from spike inputs. In biological neurons this is mediated by another complex, partially continuous process, but in SNNs a simpler mathematical formula is used [34]. Neuronal integration in this paper was computed with the Leaky-Integrate-and-Fire (LIF) neuron, a simple analytical application of membrane time constant, firing threshold and refractory period, on an abstract level similar to the mechanism observed in biological neurons [31].

3.3 Connectivity

Working with only one recurrent layer, a singular connectivity matrix of the shape `[to_neurons,from_neurons]` was set up, with zero-values representing no connections and weights inserted for neuron connections. The matrix is initialized with zeros and then can be filled either manually or with a function automatically setting random connections at a connectivity percentage of choice. The `auto_connect` function ensures that connections are relatively balanced. This means that each neuron is connected to a similar number of neurons in order to ensure the whole network is interconnected rather than forming separated sub-networks, and that each neuron gets a similar amount of inputs at full activation. Unbalanced connections would likely decrease synchrony, since optimal adaptation to other neurons relies on short pathway feedback loops. With a balanced approach like in this model, each neuron is connected to each neuron in the network through few connections, facilitating quick communication. For the random connections a seed was used for reproducibility.

Additionally an input matrix was set up with `[neurons, time_steps]`, if for example poisson inputs are supposed to be added. At default mode the input matrix is set to zeros.

For analytical simplicity, the weights of the model were kept exclusively excitatory. Although some previous work has suggested that a balance of excitatory and inhibitory connections is favorable when trying to achieve synchrony, other recent studies show that inhibitory connections are not a hard requirement for synchrony [14].

3.4 Voltage Input

In modeling spiking networks it is common practice to simulate input from another connected neuron population or randomly drawn from a poisson distribution, to mimic input as observed in biological neuron populations [1, 7, 14]. For this model the spike input matrix was set up exclusively as an option to test code functionality and to support debugging processes. In the main experiment, input to neurons was given solely in the form of noise and lateral connections. The purpose of refraining from additional input was to isolate the noise effect, in order to test the experimental hypothesis.

3.5 Integration

In the integration step of the code, the voltage is updated per neuron and time step by the LIF calculation. The formula takes input currents and noise of the previous time step and calculates a membrane voltage update. The function then checks for the refractory period, where for all cases that are not in refractory, membrane voltage is compared to the threshold voltage and then updated. In the case that the neuron spiked after the voltage update, the last step is to reset the membrane voltage to the resting potential. Each of these steps is performed along the full neuron vector for one time step in at the same time, because vector calculations are highly computationally efficient compared to loops. Code from Zemliak et al. [36] was used as a template to build this structure and adapted according to the simplified experimental use case. In the LIF formula (see 11.1), voltage calculations modified by input currents are divided by the product of delta time and membrane time constant. Delta time represents the step size operated on, here set to 1 ms steps for simplicity. Analytically, the membrane time constant is the product of membrane capacitance and membrane resistance, which describes the rate at which the membrane rises or falls in biological neurons. The membrane time constant was set to 10, which lies within the biological range [20].

3.6 Simulation

The *simulate* function takes care of running through each time step of the spiking neural network simulation. Voltage and spikes are recorded in separate matrices after each integration time step. The integration function inside the time steps loop is fed by wih a matrix of external and lateral input currents combined by a dot product. External input represents potential input streams from outside the networks, which was set to zero in the final experiment of this paper. Lateral input stems from the directed neuron connections inside the network. THese inputs are zero for connections where preceding neurons did not fire in the previous time step and larger than zero for connections where preceding neurons did fire. Noise is added inside the integration function instead of combined into the input current matrix. Again, code from Zemliak et al. [36] was used as a template to build this structure and adapted according to the simplified experimental use case. For more accurate visualization another voltage matrix was added in this paper, where voltage was recorded twice per time step, once at the point where input currents have recently changed the voltage, and once after a potential spike. In the case of a spike this leads to two different voltage recordings, as the voltage is reset after the spike. Without this double recording the voltage increase right before a spike will be missing from visualizations.

4 Hyperparameter Setup

4.1 Resting Potential and Firing Threshold

Resting potential and firing threshold in biological neurons are not identical among neurons, as membrane channel configurations differ depending on the neuron's functionality. The firing threshold can be influenced by spike dynamics as well as intrinsic noise and can thus vary even within a singular neuron [21, 24]. However, for voltage dynamics to allow for spikes, resting potential has to lie in a certain range, depending on the neuron type between -30 and -100 mV [20]. For this model resting potential was set to -65 mV, the mean of the biological range. The firing threshold was chosen according to the typical value in biological neurons, which lies around -55 mV [25], which was also used in the model of Zemliak et al. [36].

4.2 Refractory Period

The parameter *refractory period* is composed of two factors, the absolute and relative refractory period. In biological neurons, the absolute refractory period refers to the time frame where it is physically impossible for another spike to be fired, because the current spike is still in a depolarization or early repolarization stage, where all sodium channels in the membrane are closed. Furthermore, allowing for voltage increase in this phase would eliminate the binary nature of spikes. The relative refractory period begins once sodium channels start to recover and potassium channels begin to open for hyperpolarization. This state makes it unlikely to evoke a new spike, but possible in case of very strong input stimuli [20]. In theoretical models, absolute refractory period is implemented at the step of calculating spike summation for the next spike, which can be set prior to running the model. A relative refractory period is usually not explicitly implemented by simpler models, as they utilize a reset to resting potential and disregard hyperpolarization. However, it can be mimicked by the slow rise of sub-threshold voltage from resting potential towards the threshold. As long as the voltage is still close to the resting potential, only strong inputs can lead to spikes, whereas once it gets closer to the threshold, spikes are evoked easily.

In the first trial runs of this model, absolute refractory period was set to zero, due to the assumption that the slow rise in voltage would act as relative refractory period adding a strong enough refractory effect for this experiment. After the fine-tuning of several hyperparameter values, the absolute refractory period in the main experiment of this paper was set to 3 ms, minimally longer than the average biological range of 1-2 ms, as it turned out to improve visual pattern distinction.

4.3 Noise

The noise in this experiment serves to imitate sub-threshold activity of a neuron. Consistent with biological findings that membrane potential fluctuations can be approximated by a Gaussian distribution, the noise in this experiment was drawn from a Gaussian distribution [7]. Random noise values are added to the voltage input that is calculated in each LIF integration step, with different values per neuron and time step. The experiment was run with noise populations of different Gaussian parameters and the results were compared and analyzed. The range of mean and standard deviation (SD) combinations was chosen to fit the experiment reasonably: The overall heuristic for experimental fit was for the noise to shift membrane voltage of each neuron closely below the firing threshold. This relates to subthreshold activity observed in biological neurons [1]. To test edge cases, this heuristic was partially loosened. Although the mean of the Gaussian noise was always set to be above zero, small amounts of below zero noise were possible due to a partially high Gaussian standard deviation. In biological neurons negative noise currents are indeed observed, because of inhibitory sub-threshold activity [1]. However, the shift of the Gaussian bell carried out in the different experimental conditions resulted in an imbalance, where some conditions would receive negative input and some would not. Therefore non-clipped Gaussian noise was discarded in the fine-tuning process and experiments were carried out with clipped below zero noise.

4.4 Number of Neurons

The number of neurons was chosen to fit the scope of the project instead of attempting to replicate any specific biological neuron cluster. For better overview and easier testing in the early stages, the number of neurons in this project was set to 10. Such a small number of neurons is limited in its informative value, because outcomes may occur due to coincidence rather than a pattern resulting from connection dynamics [12]. The higher the number of neurons, the more informative the results and the smaller the connectivity percentage necessary to yield a similar outcome. For larger amounts of neurons, the percentage of connection per single neuron can be decreased to achieve a similar global network activity and associated effects. Although the goal for increased informative value pushes towards a higher number of neurons, the choice was also limited from the top by computational power and aim for visualization readability. The final experiment was carried out with 50 neurons, with a global connectivity of 5%. For comparison: A preliminary version of the network with 10 neurons showed similar activity with a connectivity percentage of 13-15% per neuron. A network of 100 neurons was tested in the process but visualization for this network turned out and the specific experimental outcome did not differ largely from the observations of a 50 neuron spiking neural network.

4.5 Time Steps

The number of time steps was adjusted to the needs of the experiment according to two main requirements:

(1) The amount of time steps should be sufficient for the membrane potential to reach the threshold multiple times in the majority of noise configurations.

(2) Encourage spike patterns of different lengths to form, shift and stabilize. The main experiment was run for 2000 time steps, to give the network time to stabilize its patterns and to ensure the simulation length did not affect the outcome.

5 Experimental Setup

5.1 Experimental Design

The experiment was set up as a $11x11x3$ factorial design. The first two factors are eleven different selected values of mean and standard deviation for the Gaussian noise distribution, to study the effect of noise. A global weight value of neuronal connections (same weight for each neuron in the network but varied per experimental run) was added as a third factor for network variability, to ensure better reliability of observed effects.

5.2 Initial Condition

For the first time step of the experiment, all neurons are set to different start voltages within a 2 mV range around the resting potential. This was done to ensure closer biological plausibility even at the initial condition, which aims to mimic a randomly observed biological system, where even neurons measured at resting potential are expected to show small variations [7].

5.3 Experimental Measures

Four different measures were utilized to tune the hyperparameters and examine the validity and results in the experimental condition. For this process *average spike count* and *average ISI* were recorded. In the experimental condition, synchrony was measured with *Rsync* and an *Average Spike Pattern Length* measure was set up, with the intention to approximate the time a pattern takes to finish, i.e. the time steps until the whole laterally connected network has been activated and run through its ensemble (see 11.3).

6 Hyperparameter Tuning

Since the experiment has multiple interacting parameters, some hyperparameter tuning and selection had to be carried out experimentally. Few parameters were set according to literature or technical constraints, which include the number of neurons, the refractory period, number of time steps, noise distribution, as well as resting potential and firing threshold. The following parameters were selected through a careful procedure: The range of mean and standard deviation values, the global interconnectivity percentage of neurons and global weight connections between individual neurons.

6.1 Noise Range

Noise range was selected first, because of its independence of neuron connectivity in this model, in contrast to connectivity, which is affected by the noise input. For the noise parameter to be of experimental interest, it had to fulfil at least one of two criteria:

(1) Push membrane voltage of neurons towards values that lie closely below the firing threshold, a behavior observed in biological neurons [1].

(2) Offer the possibility to test edge cases.

Voltage was modeled in a tuning experiment where the network was neither interconnected nor provided with a feedforward drive, in order to observe isolated noise behavior for choosing relevant parameters. The range for both mean and standard deviation variables was selected to be 0.1 to 1.1 with these justifications:

6.1.1 Gaussian Mean

a) For low means such as 0.1, (1) was mostly not fulfilled, unless combined with high standard deviation, thus fulfiling (2).

b) For means of around 0.8, criterion (1) was fulfilled.

c) Means higher than 0.9 push voltage of some neurons over the firing threshold, thus partially but not completely failing to fulfil (1), and also staying relevant for (2). 1.1 was roughly estimated to be the last value to still fulfil (2).

d) Means above 1.1 were excluded because most or all neurons would fire constantly from the noise raising voltage above the threshold.

6.1.2 Gaussian Standard Deviation (SD)

a) Low SD values such as 0.1 fulfilled (1) when combined with a high mean.

b) SD values above 0.5 were identified to support low means in reaching voltage values close to the threshold, thus necessary to fulfil (1) in some neurons.

c) 1.1 was selected as the top limit for SD, although higher values still fulfilled (1) for some neurons when combined with a low mean. The upper limit was decided due to the estimate that higher values would not make a substantial difference and because the range had to be limited to fit the scope of the experiment.

6.2 Network Connectivity

Neuronal connectivity percentage in biological networks varies depending on the size, location, function and neuronal plasticity of the assembly. The same applies to individual connectivity weights [35]. Since the model in this paper is simplistic and not built after a certain observed biological network, connectivity values cannot be selected from biological backgrounds, but have to be adjusted to fit this specific model. The first of the two variables to be considered is the global connectivity percentage, which describes the percentage of neurons from the whole network that each individual neuron is connected to. The second variable is the strength of each interconnection between individual neurons. In the experiment the weight variable is set to a universal value, meaning each lateral connection has the same weight per experimental condition. A universal weight value ensures simplicity and reliability of this experimental variable; however, it has to be noted that this is not biologically plausible [9, 35]. Since the two connectivity variables interact, tuning has to be performed simultaneously; first, different arbitrary combinations were tried, until values offered readable results, i.e. neither over- nor under-stimulation with mean noise conditions. Then, global connectivity percentage was manipulated while the individual weight value was kept fixed, because the former is more sensitive to changes. At this stage of the tuning process, individual connections weights were set to 30. Values were then tuned according to two criteria: (1) Find the lowest combination of both variables to still produce a reliable pattern and (2) choose the lowest global connectivity percentage to still connect all neurons to form one coherent assembly, strictly preventing any sub--networks and disconnected neurons.

6.2.1 Global Connectivity Percentage

The global connectivity percentage was evaluated by looking at the connection graph of the 50 neuron network. Figure 1 shows visual examples of different connectivity percentages and their connection density.

a) For a global connectivity percentage of 0.01 criterion (2) was not fulfilled, where only around half of neurons formed any connection.

b) For a value of 0.03, connectivity was still too sparse, with many neurons forming only a single connection, thus not conforming to (2).

c) A percentage of 0.04 equipped some neurons with only a singular connection, which technically just about fulfilled (1) and (2), but was disregarded to be sure the network was balanced and running smoothly.

d) A percentage of 0.05 lateral connectivity in the 50 neuron network was the lowest value to reliably produce a fully connected network with each neuron forming at least two connections, while appear relatively balanced. This value was chosen for the experiment, because it fulfilled both criteria.

6.2.2 Uniform Connectivity Weigths

Uniform global connection weights were selected while looking at the average spike count, a good measure for lateral input efficacy. Higher spike counts correlate with stronger lateral connectivity, since neurons get higher input and thus are more likely to spike. At the same time biases are avoided for the critical dependent variables, which are synchrony and spike pattern length. To make sure that effects of lateral connectivity outweigh noise effects, the minimal requirement for the lowest universal weight value was expected to (1) increase spike count by a factor of 1.5 compared to the spike count observed with solely noise input. From the lowest universal weight value fulfilling criterion (1), two higher values are supposed to be selected that fulfil criterion (2) that the step size between the values results in at least a 10% increase in spike count. The choice to limit this experimental variable to only three values was made to stay within a reasonable scope for this project.

a) Universal weights below 20 mV did not fulfil (1), so they were disregarded. 20 mV was set to be the lowest experimental condition.

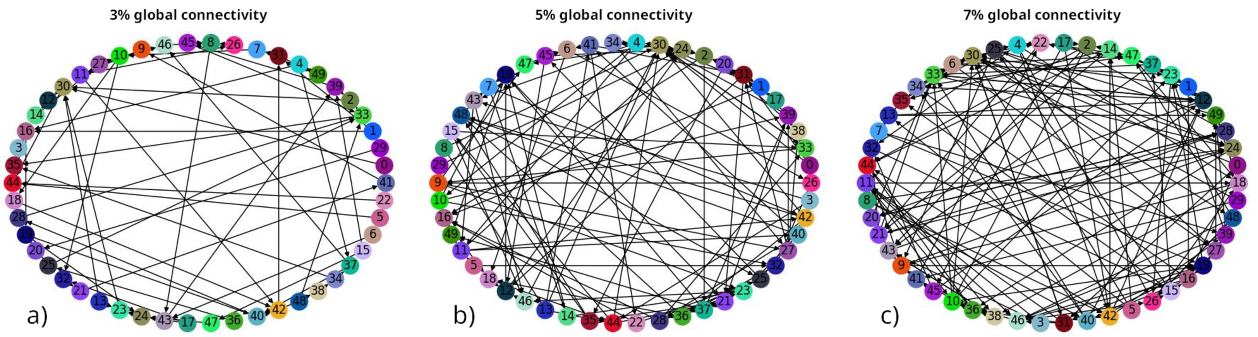

Figure 1: Laterally Connected 50 Neuron Network

b) For the following two values, an increase of about 15% in average spike counts was observed with a step size of 10 mV. The values 30 mV and 40 mV, fulfilling criterion (2), were selected as the remaining two values for the universal weight variable.

6.2.3 Overview

The completed tuning process resulted in these parameters:

- Gaussian mean: 0.1-1.1 (step size 0.1)
- Gaussian SD: 0.1-1.1 (step size 0.1)
- Global connectivity percentage per neuron: 5%
- Universal lateral connection weights: 20, 30, 40 mV.

It has to be noted that a seed was set for reproducibility, thus the experiment did not need to be run more than once in the same experimental condition.

7 Predictions

In this paper I hypothesize that (1) synchrony appears with an optimal amount of non-zero noise and that (2) synchrony correlates with a shortening in pattern length due to its parallelization effect. Different noise configurations are expected to yield different results:

a) Low noise amplitude combined with low noise variance is predicted to lead to no spikes or few spikes. Since the mean is not close enough to the firing threshold, it does not induce spikes on its own. Low variance will keep values close to the mean, which may only lead to values above the threshold in few cases, where drawn values happen to be large enough. Synchrony is not expected in cases of zero or few firings. Absolute spike pattern lengths are expected to be zero or very short in the case of few spikes, however these do not represent an actual ensemble pattern.

b) Low noise amplitude in combination with high noise variance is expected to cause some spikes, which may or may not lead to ensemble patterns. Synchrony is not expected to be high with this noise configuration. Absolute spike pattern length is predicted to be long for ensemble runs and short for individual spikes fired that do not continue to activate the ensemble.

c) High noise amplitude with low variance is expected to be the optimal noise configuration for synchrony, because it is predicted to activate multiple neurons at the same time, due to values sticking to a similarly high values for each neuron. This should lead to whole-ensemble activations with parallelized action, which should shorten pattern length compared to noise conditions where fewer neurons are activated simultaneously.

d) High noise amplitude combined with high variance is predicted to lead to diverse neuronal activation. On the one hand, high and medium noise values push neurons to go above threshold, starting and continuing the ensemble pattern. On the other hand, the expectation of several very low noise values may prevent certain neurons from firing, even if the lateral connections strongly increase the voltage of those neurons. This may result in some shorter, broken patterns, but mostly longer

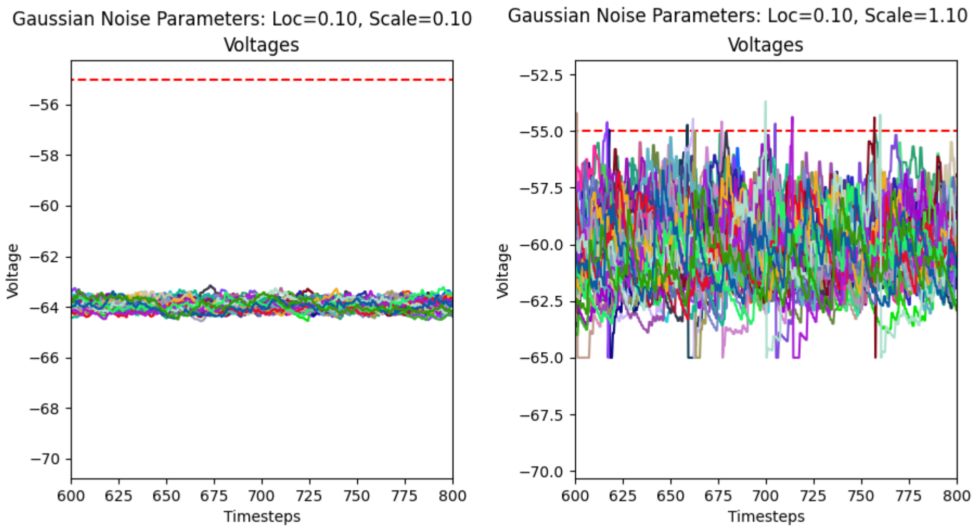

Figure 2: Voltage Recordings with Zero Interconnectivity in the 600-800ms Window

patterns running through the whole ensemble without efficient parallelization of spikes.

Results are expected to show relatively smooth transitions between conditions.

8 Results

First, spike and voltage plots per condition were analyzed, with observations from the 600-800 ms window of the 2000 ms trials, a time stamp where it was estimated that enough time had passed for noise to push sub-threshold behavior closely below the firing threshold. There was an overall trend in behavior that was correlated to the noise parameters. In the following section, any mentions of *loc* and *scale* are referring to the Gaussian distribution parameters from which noise values were drawn, with loc considered the mean and scale the variance of the distribution.

8.1 Voltage Analysis

Combinations of low locs and scales keep membrane voltage hovering closely below the firing threshold. Voltage can be observed like a visual band, where higher locs push this band towards higher overall voltage values and higher scales widen the band to span an overall broader range of voltage values. These results are intuitive considering the function of those parameters in a Gaussian distribution: the loc parameter shifts the position of the mean, while the scale parameter shifts the variance, i.e. the breadth of the Gaussian bell. Another important observation not immediately inferred from looking at the distribution is a raise in overall voltage values with increased variance but unchanged loc. This effect is caused by the clipping of below zero noise values (see 4.3). For noise configurations where the Gaussian bell is shifted further into the negative, but then negative values are discarded, higher variance values are drawn in their place and thus the likelihood for higher noise values overall increases. As seen in Figure 2, where noise was isolated from interconnected neurons (interconnections set to zero), at a loc of 0.1 and scale 0.1 the voltage mean lies around -64 mV, whereas for loc 0.1 and scale 1.1 it rose to about -60 mV, with gradual transition steps for in between scale values. For higher locs this is not observable, because even noise isolated from interconnections leads to frequent or continuous spiking, resulting in a skewed voltage band caused by the voltage resets after spiking, thus is not representative of voltage fluctuations caused by noise exclusively anymore.

8.2 Spike Pattern Analysis

Across connectivity conditions, low loc and scale configurations usually lead to no spikes. Increasing either of the values results in an increase in spikes.

For low locs and increasing scales, first few spikes appear that are scattered across neurons (see Figure 3a). Such spiking behavior does not lead to patterns, because spikes are too sparse. Sparseness despite strong interconnectivity is caused by overall low noise, which is unable to push voltages towards the threshold. Some neurons may receive higher noise input and spike, however neurons that receive this spike as input are still likely to stay below the threshold, because their voltage is still too low to spike. Thus the ensemble breaks down after one or few spikes, preventing full patterns from being carried through. This results in most parts of the ensemble to neither be triggered by interconnectivity, nor by the noise. As to be expected this effect varies between interconnectivity conditions: For the same noise conditions, higher interconnectivity values result in a longer spike chain, because it is more likely for the next components of the ensemble to be pushed over the threshold if their combined input (noise + weights) is higher (see Figure MISSING).

For low locs and increasing scales, patterns begin to emerge visually. At first these patterns appear partially broken and not periodic, spikes still look more scattered than ordered (Figure 3b). Transitions between *no spikes → few spikes → broken patterns → periodic patterns* shift depending on the noise configuration as well as the interconnectivity weights. Especially with smaller interconnectivity weights, even for increasing locs and arbitrary scale values separated patterns do not emerge. For interconnectivity weights of 20mV the first clearly distinguishable pattern appears with a loc = 0.7 or loc = 0.8 (Figure 3c), while for an interconnectivity weight value of 40mV patterns are distinguishable for the first time at loc = 0.3. Clear patterns appear when the interconnectivity effect outweighs the noise effect. For higher interconnectivity weights, patterns are more quickly separated, because the chance for all neurons in the ensemble to be activated is higher even at lower noise amplitudes. For example, the assembly with a 40mV weight per connection essentially gets double as much interconnectivity voltage input as the assembly with 20mV connection weights. Thus connected neurons are much more likely to fire after their preceding connections have fired, even with lower sub-threshold activity.

Further observations can be made about the distance of patterns once they emerge. For certain noise and connectivity configurations, patterns appear clearly separated. For 20mV interconnectivity, the first clearly separated full pattern appears at loc = 0.8 and scale = 0.2. This interconnectivity condition already shows emergence of patterns at lower locs but those do not appear clearly separated and potentially incomplete (Figure 3b). Once patterns have reached a loc value where they appear clearly separated, with increasing scale they then begin to move closer together until they appear to overlap (Figure 3d). For clearly separated patterns it appears that the higher the loc, the shorter the pattern length. These observations can be made about all interconnectivity conditions, but the onset of clearly separable patterns occurs with different noise configurations. The higher the interconnectivity weight, the earlier the onset of clearly separable patterns. For the connectivity conditions it is visually indistinguishable whether pattern size changes with increasing scale but consistent loc. [Both patterns and pattern intervals displayed consistent behavior in length within individual experimental conditions ??? MISSING sense]

8.3 Synchrony Analysis

Next, spike behavior and synchrony were put into context: For this, spike plots were visually analyzed in the 600-800 ms window and related to the corresponding Rsync values (see Figure 4). For all interconnectivity conditions, zero spikes result in zero Rsync and few spikes result in very small Rsync values, usually below 0.05.

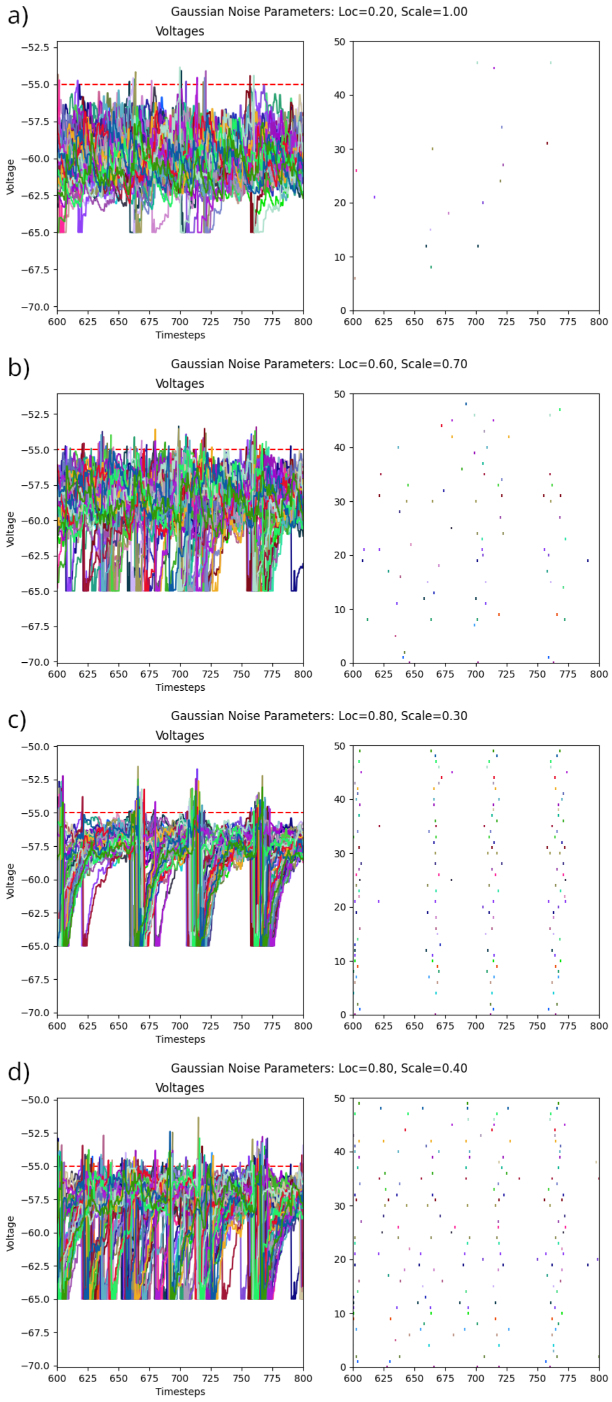

Figure 3: Spike Behavior in Different Noise Conditions and Corresponding Voltages for 20 mV Connectivity Weights

Increasing loc and scale leads to more spikes. For interconnectivity weights of 20 mV, loc = 0.4 with scale = 1.0 correlates increased spikes, but Rsync stayed below 0.05, because spikes are still very scattered without visible patterns. This reflects a state where effects of interconnectivity are still smaller than the noise effect. At loc = 0.7 with scale = 0.5, the 20 mV weights condition first reaches an Rsync value above 0.05, with visual spike trains slowly separating, but reliable full ensemble runs not detectable yet. Loc = 0.7 produces some Rsync values over 0.05, correlating with incomplete patterns, but Rsync quickly subsides again with increasing scales: The decrease in synchrony after a peak at scale 0.6 and 0.7 correlates with previous incomplete spike patterns beginning to overlap.

At loc = 0.8, scale = 0.2 a first onset of higher synchrony (Rsync = 0.21) was recorded. This correlates with first clearly separable and complete ensemble patterns. Synchrony steadily dissipates again after a certain peak (loc = 0.8 at scale = 0.3), at the same time as patterns have moved as closely together in time to overlap. At this point, even once synchrony decreases with higher scale, it does not fall below Rsync of 0.05 until scale = 1.0, where patterns become inseparable. For this interconnectivity condition a global peak of synchrony (Rsync = 0.59) was reached at loc = 0.9 and scale = 0.1, with the visually shortest patterns.

With increasing scale, patterns widen and move closer together. At loc = 0.9 and scale = 0.4, synchrony drops to a lower value (from Rsync = 0.29 to 0.14), which correlates with the between pattern interval drastically shrinking. Following scales at the same loc show the begin of overlapping patterns. At loc = 1.0 and scale = 0.1 synchrony is still comparably high (Rsync = 0.44) but lower than for the previous loc at the same scale value (Rsync = 0.59). This decreased synchrony is reflected in shorter between pattern intervals compared to the previous loc condition. Since 0.1 is the lowest possible scale value in this experiment, it can be assumed that synchrony would peak at even smaller scale values, but it is unclear if smaller scale values would be reasonably applicable.

For the last loc condition, synchrony begins high again (Rsync = 0.3), with the peak possibly out of range. Here patterns appear separable and very short, but with an already small between pattern time interval. Because of these short separation intervals, patterns quickly begin to merge as scale increases.

For interconnectivity weights of 30 mV, synchrony and spike behavior look different to

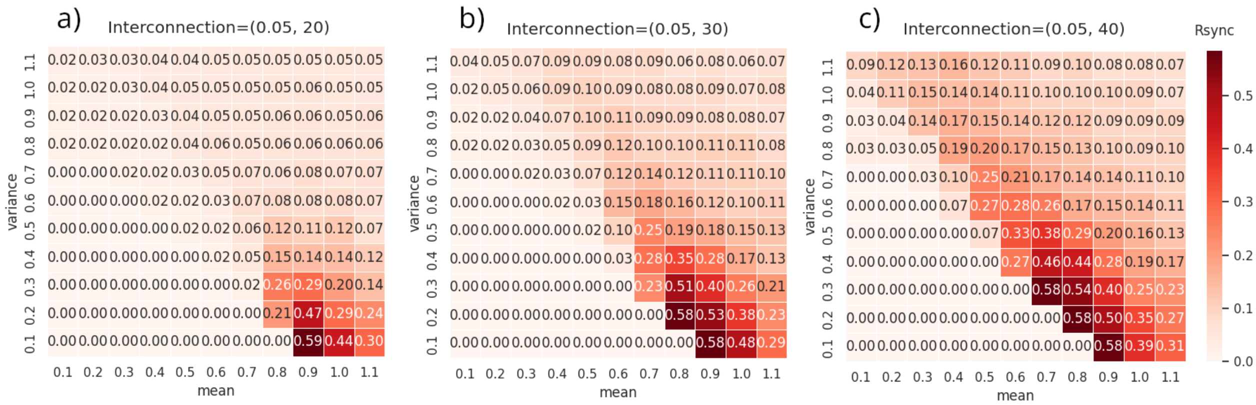

Figure 4: Rsync Results across different Interconnectivity Conditions

20 mV weights in noise conditions that produce more than a few scattered spikes. Here, rough, incoherent patterns can already reach Rsync values a little over 0.05, which can be observed for loc = 0.3, scale 1.0 and loc = 0.4, scale 0.9. For locs 0.5, synchrony appers at a certain scale level and then decreases again after a singular peak. The higher the loc, the earlier and stronger the onset of synchrony: for loc = 0.6 onset happens at scale = 0.5, peak at scale = 0.7 with an Rsync value of 0.15, and subsides gradually from scale = 0.7 upwards. For loc = 0.8 the onset of higher synchrony, which marks its peak for this loc, already appears at scale = 0.2, with a nearly quadrupled Rsync value (= 0.58) compared to loc = 0.6. Synchrony subsides from scale = 0.3 onwards with stronger synchrony decreases between scale steps. With this trend of earlier synchrony onset, it should be considered that the peaks for locs >0.9 may fall out of the experiment's recording range. This would be in line with the observation that locs >0.9 have shrinked peak values at scale = 0.1, the lowest possible scale value.

Rsync peaks correlate with the most separable of full patterns, and decreases of synchrony with higher scale values correlate with the degree of pattern overlap. Higher locs result in overall stronger synchrony and shorter full patterns, as long as separability is maintained. This effect can be observed when looking at the global peak in synchrony, which correlates with very short patterns and long between pattern intervals. Decreases in synchrony at higher scales then correlate with smaller separation intervals, where pattern length stays consistent. For loc = 1.1, scale = 0.1, patterns are very short and still separable, with an Rsync value of 0.29. Synchrony is only half as strong as the global maximum despite its short intervals, presumably because mathematical pattern separability is not as strong with smaller between pattern intervals, despite clear visual separation.

For the last interconnectivity condition of 40mV lateral weights, synchrony for zero spikes and few scattered spikes behaves the same as with the other interconnectivity condition, where zero spikes result in Rsync = 0 and with very few spikes Rsync stays below 0.05. In this interconnectivity condition the onset of synchrony appears even earlier. Loc = 0.2, scale = 1.0 already show separable but broken patterns, with a first Rsync value over 0.05. For comparison: with 20mV lateral weights, first Rsync >0.05 is reached at loc = 0.6, scale = 0.8. With 40mV, the interconnectivity effect quickly overpowers the noise effect. Observed patterns are more variable in length and separation intervals with low loc to high scale combinations than with higher locs. At loc = 0.5, scale = 0.6, Rsync first goes over 0.2, a comparably high value, with more stable pattern and separation interval lengths. Synchrony again subsides where patterns begin to merge. The global synchrony peak for 40 mV weights is recorded at loc = 0.7 and scale = 0.3, but it is notable that this global peak happens three times in total, from loc = 0.7 through loc = 0.9. With each increasing loc step the scale value for the same Rsync is one lower than for the previous. For all three global peaks synchrony subsides from the next scale value onwards, where earlier subsiding also correlates with earlier fading, with comparable step sizes. Here again higher synchrony correlates with increased separation interval length and shortness of patterns. This can roughly be estimated from the visual spike patterns in the 600-800ms window:

Table 1: Pattern and Separation Intervals with 40 mV weights

Gaussian Loc	**0.7**	**0.8**	**0.9**
Gaussian Scale	**0.3**	**0.2**	**0.1**
Rsync	0.58	0.58	0.58
Separation Intervals	>150ms	>150ms	>150ms
Pattern Length	$\sim$ 5-8ms	$\sim$ 5-8ms	$\sim$ 5-8ms
Gaussian Scale	**0.4**	**0.3**	**0.2**
Rsync	0.46	0.54	0.50
Separation Intervals	$\sim$ 15-70ms	$\sim$ 20-75ms	$\sim$ 15-50ms
Pattern Length	$\sim$ 8ms	$\sim$ 8ms	$\sim$ 5-8ms

The results in Table 1 show that a correlation of increased synchrony with shortness of patterns and separability is to be expected. So far it is inconclusive whether one of the two factors is a stronger or the main predictor. To compare the effects of separability and pattern length, specific examples were selected that were able to compare two cases:

(1) Similar pattern lengths but different separability intervals (loc 0.7, scale 0.3, 40 mV, higher separability; loc 0.8, scale 0.3, 40 mV, lower separability).

(2) Different pattern lengths but similar separability intervals (loc 0.5, scale 0.7, 40 mV, longer patterns; loc 0.8, scale 0.3, 40 mV, shorter patterns).

Those two comparisons showed that:

(1) If pattern length was similar, synchrony was observed to be higher for intervals that are further separated in time (Rsync 0.58) than for patterns with lower separability (Rsync 0.54).

(2) For similar separability intervals, shorter patterns showed higher synchrony (Rsync 0.54) than longer patterns (Rsync 0.25).

It has to be noted that these observations are very sparse, because they were done after the experimental run and most noise conditions did not serve to investigate this interaction without additional measures. Further experiments with more examples would be necessary to confirm my assumptions.

Comparing the global peaks of the 30 mV with the 40 mV interconnectivity condition, for the former, the same value (Rsync = 0.58) is first recorded at loc = 0.8, scale = 0.2, whereas for the latter it already appears with loc = 0.7, scale = 0.3. While with 30 mV weights this value is recorded twice under selected noise conditions, with 40 mV weights this value is recorded three times, most probably due to an earlier onset on synchrony. With increasing locs this peak is consistently observed with decreasing scale values, which would predict it to fall out of range from loc 1.0 onwards, after it was observed for the smallest scale value (= 0.1) for loc = 0.9. This is indeed observed for both the 30 mV and 40 mV interconnectivity condition.

8.4 Average Pattern Length Analysis

While conducting the experiment, I also recorded an average pattern length, however the results do not reflect visually estimated pattern lengths consistently well under different noise conditions. This is due to the experimental nature of the recording method as well as the strong variations of pattern lengths in some conditions, compared to other conditions where pattern length stays consistent. For completeness, results of the experimental pattern length measure are presented as well.

The results of this measure look similar across all interconnectivity conditions. For low locs and scales no pattern lengths are recorded, which correlates with the absence of spikes. With low locs but higher scales, average pattern length appears to lie between 1 and 3 ms, which correlates with few scattered spikes not yet forming coherent patterns, because overall noise is too low for interconnections to continue activating the ensemble. With increasing locs, lower scale values can already result in spikes and thus average pattern length is recorded here as well. A trend of shorter average pattern length correlating with higher synchrony can be observed, where combinations of high locs and low scales yield best synchrony results and comparably small average pattern lengths of 6 to 8 ms. With increasing scales at higher locs, average pattern length increases, with values of 10 to 20 ms. This trend is disrupted by some outliers, such as 18 ms at 20 mV interconnectivity weights at loc = 0.8, scale = 0.2, despite high synchrony (Rsync = 0.21). These outliers where synchrony is very high and pattern length is not as short as expected, can be attributed to the experimental nature of the average pattern length measure, as well as the way the Rsync measure works. Rsync does not only take the degree of spike parallelization into account, but also separability, which can be seen in the length of between pattern intervals. Those outlier values do not have the shortest patterns, but still show high Rsync, because separation intervals are very long, increasing the separability effect.

9 Discussion

9.1 Interpretation of Findings

The present paper has revealed that synchrony can emerge in a recurrent spiking neural network receiving an optimal non-zero amount of Gaussian noise input. What makes those noise parameters optimal depends on the lateral connectivity of the neuron ensemble. With lateral connectivity strong enough to connect a whole network without creating gaps or subnetworks, synchrony onset depends on the strength of connections between neurons. In a network with stronger interconnections, the onset of synchrony appears at lower noise input and the network displays higher overall synchrony values, compared to a network with weaker interconnections.

A general trend for the Gaussian noise parameter configuration points towards a combination of high mean and low variance. In order to reach optimal Gaussian noise conditions inducing synchrony, a high chance for multiple neurons to spike in parallel has to be achieved. Combinations of low mean and low variance fail to spike enough neurons, because sub-threshold behavior stays too far away from the firing threshold, thus no spikes are elicited without external input. If the variance is increased, despite a low mean parameter, some neurons may be pushed close enough to the threshold due to occasional higher noise values. However, since the voltage of most neurons is still too far below the threshold, most lateral connections will not be strong enough to cause a spiking chain reaction, failing to activate the whole ensemble pattern. If instead variance is kept low but the mean is increased, the voltage of more or most neurons will closely hover below the threshold, which then makes it possible for even small variant noise inputs to push neurons over the threshold. As soon as one neuron is activated, the interconnectivity effect is strong enough to activate the whole ensemble, creating a characteristic spike pattern representing the structure of this specific network. Depending on the mean value, one of the lowest variance values induces the highest synchrony, due to the activation of multiple neurons at the same time. Instead of running through the ensemble pattern from beginning to end, multiple parts of the pattern are activated at the same time, thus parallelizing processes. This way, the time it takes to run through the whole ensemble is shortened, decreasing the overall length of the pattern without compromising the information conveyed.

In conclusion, Gaussian noise with lower variance and higher mean values comprises the most optimal noise for synchrony induction.

When variance is increased from this optimal noise setup, synchrony slowly dissipates again, until the noise effect begins to override the interconnectivity effect. At first, Rsync values only decrease where patterns move closer together in time, due to reduced information distinction. This reduction in separation interval length is caused by an increased probability for higher than average noise values, inducing more frequent ensemble runthroughs. When variance is set even higher, patterns are pushed even closer together in time until they merge and synchrony dissipates. Under such noise conditions neurons still fire simultaneously, but the amount of firings begin to overlap so much that they lose their ability to discern information, and hence the coding quality of synchrony gets lost in information overload. Singer [29, p. 55] calls this kind of issue a "superposition problem".

9.2 Implications

The findings in this study pose important implications for research on synchrony as a coding mechanism. Firstly, the induction of synchrony in a laterally connected neural network through noise shows the internal coordination capabilities of simple neural network feedback loops. It opens up the possibility for synchrony as an internat neuronal mechanism occurring with or without external input. In addition, the synchrony-inducing effect of noise highlights that biological neural networks, which are exposed to high amounts of noise due to the nature of their environments, may not just reliably work in spite of, but because of said noise. The findings in this study are in line with the results of Manjarrez et al. [17] and Pisarchik et al. [24], which found that an optimal amount of non-zero noise promotes the emergence of synchrony. From these results it can be assumed that the brain is capable of much richer information processing than previously predicted and that not all present noise is simply filtered out.

Secondly, the observed shortening of patterns demonstrates the capability of spike synchrony to increase information coding efficiency by parallelizing processes. Synchrony thus shows strong potential as a coding mechanism because of its high efficiency and capability to encode distributed information, as has been shown in other literature [13, 36]. It is expected to operate as a complementary mechanism to rate code [29, 36].

The results in this study could inspire future research to further investigate the underlying mechanism of synchrony emerging from noise. Promising insights may be derived from dynamical systems theory, where theories of the brain modeled as a dynamical system could reveal that neural pocessing may rely on noise for the emergence of synchrony [33]. Advancing theories about the emergence of synchrony through noise could then support the development of experiments to test whether this mechanism of noise-induced synchrony can actually be observed in biological neuronal networks.

9.3 Limitations

Because of the simplicity of this study, further research is needed on more biologically plausible models to confirm the shortening of patterns linked to increased synchrony. The model in this study was constructed to resemble a biological system as best as possible, however, due to the scope of the project, some variables were strongly simplified. One of these simplified variables are interconnection weights, which were set to a uniform weight value for each neuron in the network. In future experiments such lateral weight values should be drawn from a distribution to mimic the variability in connection strengths that biological neurons display [21]. Furthermore, the network should be implemented with a learning algorithm and assigned a task, to provide evidence that pattern shortening correlating with synchrony is not solely produced under artificial circumstances, but also appears in actual applications of the network. A good example of models studying synchrony in task application are Zemliak et al. [36] and Korndörfer et al. [14], both investigated synchrony as a measure of familiarity. This type of task is particularly interesting in combination with a Spike-Time-Dependent Plasticity (STDP) learning algorithm, as it utilizes a form of Hebbian learning, a mechanism where synchrony is harnessed for Long-Term Potentiation (LTP) and Long-Term Depression (LTD) [34, 36].

Along with applied models of synchrony, more experimental evidence needs to be gathered to prove that the efficiency of synchrony code is actually what is harnessed by biological systems. While synchrony code has previously been found in biological neural networks, it still needs to be further distinguished which functions synchrony serves compared there, to the functions shown in research models.

It also needs to be further investigated whether separation intervals or pattern shrinkage are the stronger predictor of synchrony. Since the hypothesis in this paper was focused on testing the effect of synchrony on pattern length, due to the targeted process resulting in a sample size inappropriate for testing the prediction strength of separation intervals, in cannot be fully concluded that pattern shrinkage stands in stronger correlation with synchrony that separation interval length.

A further limitation of this study poses the experimental nature of the pattern length measuring techniques. Average Pattern Length was measured in a very simplified way, capturing the time window where at least one spike was recorded across the entire network per time step. However, this method does not check whether the spike sequence was caused by an actual connectivity pattern and if exactly one full pattern was counted. Further pattern length analysis was carried out on spike train visualizations, which only allows for approximations instead of precise measurements. Different measures need to be developed to determine pattern length more accurately.

10 Conclusion

This study has shown that synchrony takes distributed neural signals into account and coordinates them. With increased synchrony, more signals can be processed at the same time, which results in shorter overall processing time. Furthermore, the experiment provides an example that synchrony can emerge from an optimal amount of non-zero noise without external input, in a laterally connected neural network. Such optimal noise promotes the separation and compression of spike patterns, whereas less optimal noise fails to activate the whole pattern or moves patterns too closely together, where information gets lost due to overlap.

From these results it can be assumed that synchrony is a viable coding scheme in neurons and that it can be harnessed to efficiently process information that other coding schemes struggle to interpret. This applies specifically to distributed and time-sensitive information, which loses its informative value when converted to the more commonly investigated rate code. It was also displayed that synchrony code is much more efficient in propagating information, because of its parallelization capabilities.

It can be expected that synchrony code will become increasingly important in future research, because of its multiple advantages over rate code for certain tasks. Its ability to coordinate neural networks on a multi-cell basis is a crucial novel research area for understanding how complex thoughts as we experience them are built from the interplay of neuronal signals on the basis of small units.

11 Methods

11.1 Spiking Model

For the spiking model the Leaky Integrate-and-Fire (LIF) model was chosen, a state-of-the-art model for spiking neural networks [34]. This model is inspired by electronic principles and allows taking membrane time constant, firing threshold and refractory period into account for the integration. It provides the advantage of simulating logarithmic external stimulus to neural output relationship, which is observed in neurons[31]. The formula used in this paper was taken from Zemliak et al. [36]:

$$\tau_m \frac{dV_i}{dt} = -(V_i - E_L) + RI_i(t) \tag{1}$$

where dV is the membrane potential update with a step size of dt in ms, with the time constant τ in ms. V concerns the current voltage of the neuron, E_L describes the resting potential and $RI_i(t)$ are the incoming synaptic currents. R represents a finite leak resistance of the cell membrane [10, 36].

11.2 Synchrony Measure

To record synchrony, i.e. the degree to which neurons fire together, the Rsync measure was selected. The advantage of Rsync over other synchrony measures is its scalability to any high neuron population sizes. It works by averaging the population mean variance by the average individual neuron variance. Through this mechanism, independently firing neurons push Rsync towards 0, whereas synchronous firing averages Rsync towards 1 [36]. The Rsync formula was taken over from Zemliak et al. [36]:

$$R_{sync}(P,T) = \frac{\hat{\mathrm{Var}}[\langle A_i(t)\rangle_{i\in P}]_{t\in T}}{\langle \hat{\mathrm{Var}}[A_i(t)]_{t\in T}\rangle_{i\in P}} \tag{2}$$

P represents the encoded pattern of the neuron population and T the time interval for measuring in ms. A_i describes an activation trace of a neuron i that is part of the population P. Activation traces become more similar with increasing synchrony, and would all be identical with perfect synchrony [36].

11.3 Average Spike Pattern Length

Average Spike Pattern Length is an experimental measure developed in this paper with the intention to make pattern length analysis comparable. For this method the total spike count across the whole neuron population was taken and each time step with at least one spike count was recorded as part of the pattern. Each coherent sequence of time steps with at least one spike was summed together as one pattern to yield the amount of time steps of one pattern, i.e. the pattern length. Any amount of empty time steps were defined as between pattern intervals or separation intervals, but were not recorded in length. Of the recorded pattern lengths per experimental condition a mean was taken, resulting in the Average Pattern Length Value. The choice to already count a singular spike as part of a pattern instead of selecting a threshold of at least a few multiple firings was made because it was expected for even one spike to have value in the pattern. Due to the strong interconnectivity of the network, it was expected for most singular spikes to activate further neurons in the next step, so they were more likely to be informative than not.

It has to be noted that this measure is very limited in its informative value, because depending on the noise and interconnectivity condition, pattern length was more similar or more variable. It served to point towards a trend, but to ensure detailed and robust results, visual pattern analysis was considered more informative over the Average Pattern Length Value.

11.4 Model Connectivity

To fit the scope of the experiment, a laterally connected 50 neuron network was set up with a 5% connectivity value, which translates to each individual neuron in the network being connected to approximately 5% of other neurons. The connectivity was set semi-randomly, with a random draw of connections but a rule for upper limit of incoming connections, in order to ensure balanced lateral connectivity. The weights of the network were set to a uniform value for the whole network, only differing this global weight value for different interconnectivity conditions. The network shown in Figure 5 was used for all experimental runs.

12 Code Availability

The code of the experiment and the experimental plots are available at https://github.com/Mini11e/spiky_project.git.

13 Declaration of AI Use

AI was explicitly not used for writing in any circumstance, unless in the background of

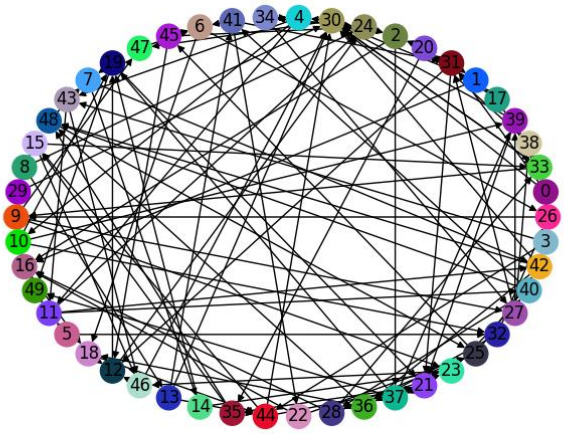

Figure 5: Laterally Connected 50 Neuron Network

browser applications for synonym and word search. For Google searches the AI assistant answers were occasionally taken as general input, but never used as a credible sources for any content written in this paper. For some LaTeX code issues, AI was harnessed to debug or adapt code to improve formatting. In the beginning of the building process for the code of the experiment, AI was used once to create a small and simplistic model to gain an overview of what a model may be structured like. This model was completely discarded from the actual experiment, but the file can still be found in the codebase under _first_version_AI.

14 Acknowledgements

This could be your acknowledgement blep.

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Contents

15 Appendix

Figure 6: Rsync, Average Spike Count, Average ISI and Average Pattern Length Recordings for 20 mV Interconnectivity Weights

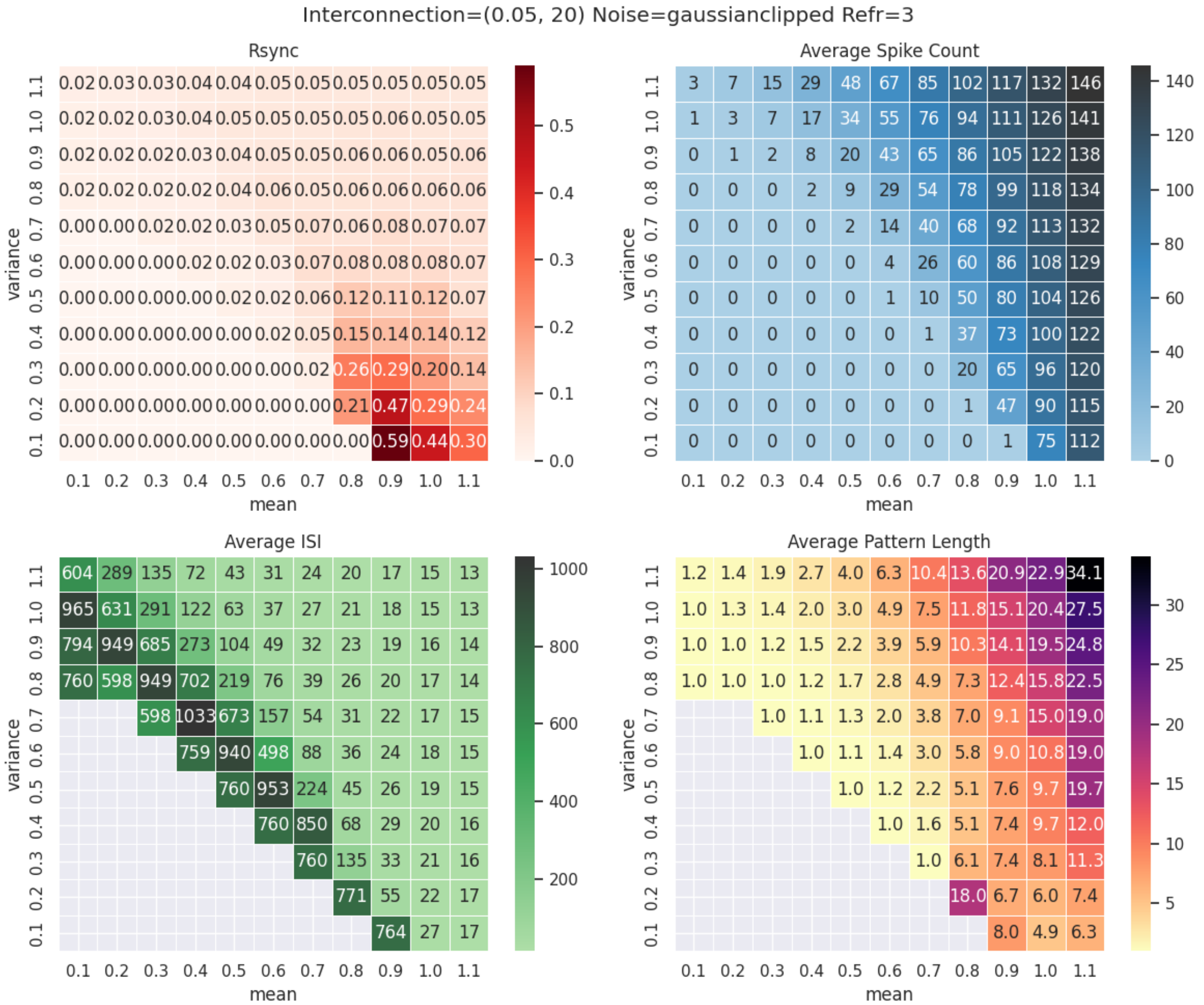

Figure 7: Rsync, Average Spike Count, Average ISI and Average Pattern Length Recordings for 20 mV Interconnectivity Weights

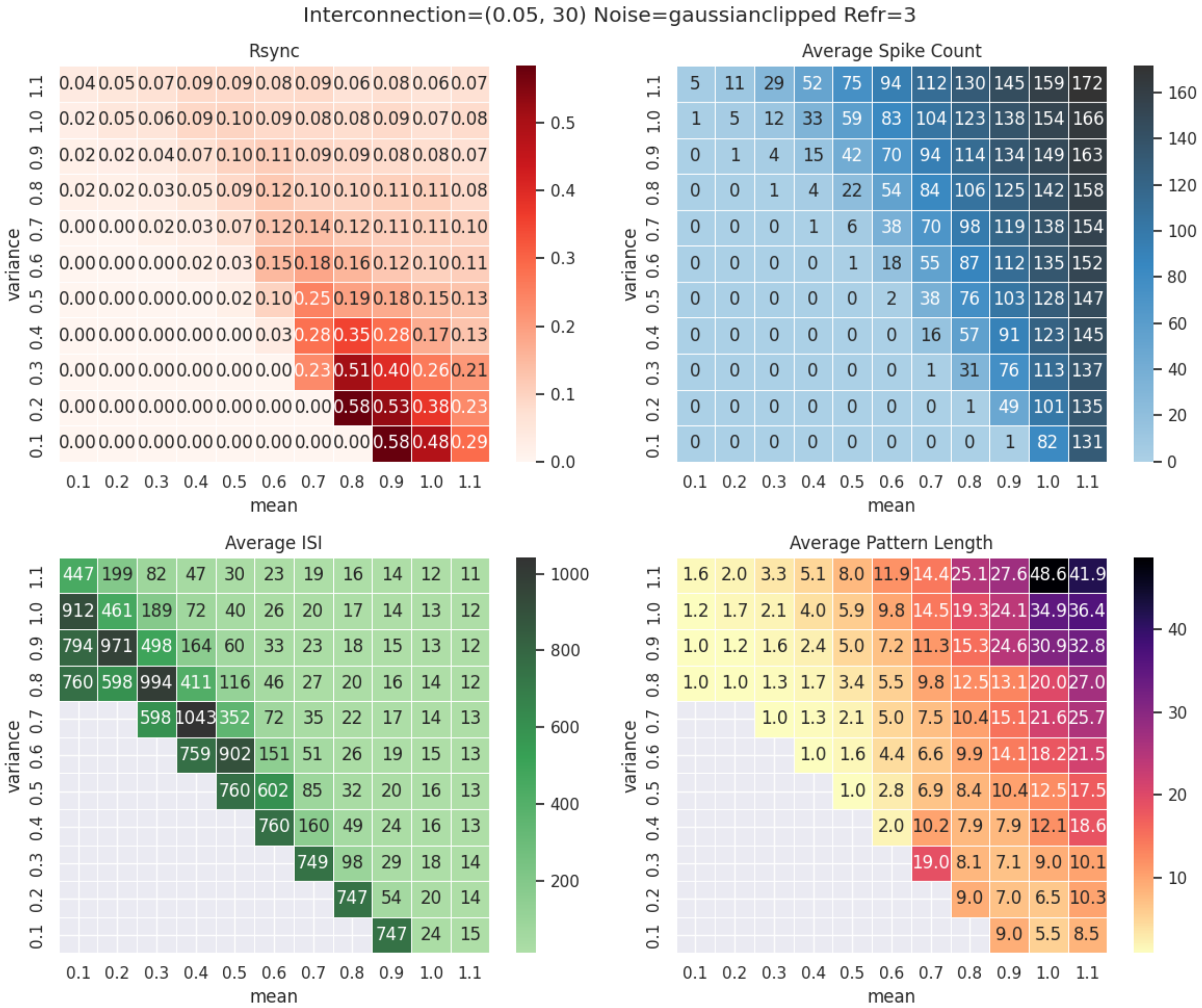

Figure 8: Rsync, Average Spike Count, Average ISI and Average Pattern Length Recordings for 20 mV Interconnectivity Weights